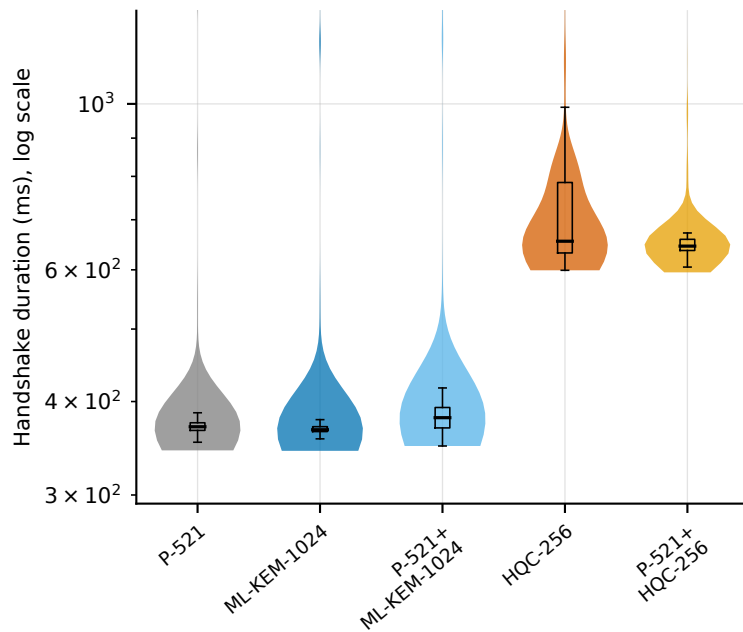
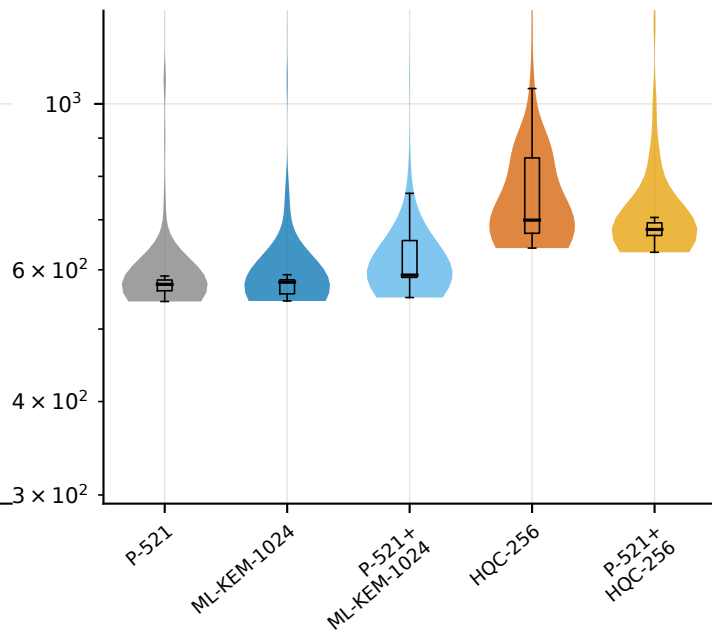


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	390.4	35.1	1.41k	605.9	26.3	2.09k
ML-KEM-1024	393.0	39.3	1.57k	609.5	30.3	1.94k
P-521+ML-KEM-1024	422.1	47.2	1.58k	634.4	27.8	2.11k
HQC-256	727.5	31.5	2.20k	789.4	31.1	2.27k
P-521+HQC-256	667.4	21.4	2.36k	735.6	28.7	3.00k